

First 5 Rows

	PassengerId	Survived	Pclass	\
0	1	0	3	
1	2	1	1	
2	3	1	3	
3	4	1	1	
4	5	0	3	

	Name	Sex	Age	SibSp	\
0	Braund, Mr. Owen Harris	male	22.0	1	
1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	
2	Heikkinen, Miss. Laina	female	26.0	0	
3	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	
4	Allen, Mr. William Henry	male	35.0	0	

	Parch	Ticket	Fare	Cabin	Embarked
0	0	A/5 21171	7.2500	NaN	S
1	0	PC 17599	71.2833	C85	C
2	0	STON/O2. 3101282	7.9250	NaN	S
3	0	113803	53.1000	C123	S
4	0	373450	8.0500	NaN	S

Dataset Shape
(891, 12)

Columns

Index(['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age', 'SibSp',
 'Parch', 'Ticket', 'Fare', 'Cabin', 'Embarked'],
 dtype='object')

```
Data Types
PassengerId    int64
Survived       int64
Pclass         int64
Name           object
Sex            object
Age            float64
SibSp          int64
Parch          int64
Ticket         object
Fare           float64
Cabin          object
Embarked       object
dtype: object
```

```
Missing Values
PassengerId    0
Survived       0
Pclass         0
Name           0
Sex            0
Age            177
SibSp          0
Parch          0
Ticket         0
Fare           0
Cabin          687
Embarked       2
dtype: int64
```

Summary Statistics

	PassengerId	Survived	Pclass	Age	SibSp
count	891.000000	891.000000	891.000000	714.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008
std	257.353842	0.486592	0.836071	14.526497	1.102743
min	1.000000	0.000000	1.000000	0.420000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000
50%	446.000000	0.000000	3.000000	28.000000	0.000000
75%	668.500000	1.000000	3.000000	38.000000	1.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000

	Parch	Fare
count	891.000000	891.000000
mean	0.381594	32.204208
std	0.806057	49.693429
min	0.000000	0.000000
25%	0.000000	7.910400
50%	0.000000	14.454200
75%	0.000000	31.000000
max	6.000000	512.329200

Median

PassengerId	446.0000
Survived	0.0000
Pclass	3.0000
Age	28.0000
SibSp	0.0000
Parch	0.0000
Fare	14.4542

```
In [16]: # =====
# DATA CLEANING
# =====

df['Age'].fillna(df['Age'].median(), inplace=True)
df['Embarked'].fillna(df['Embarked'].mode()[0], inplace=True)

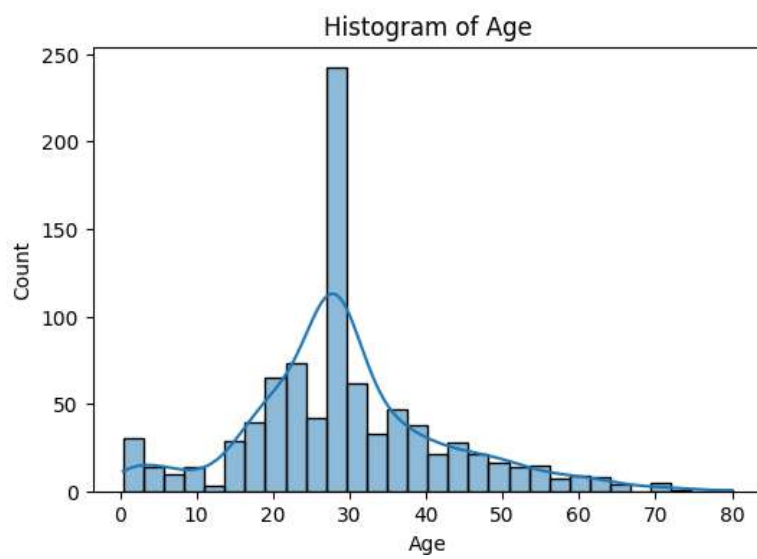
# Cabin has too many missing values
df.drop('Cabin', axis=1, inplace=True)

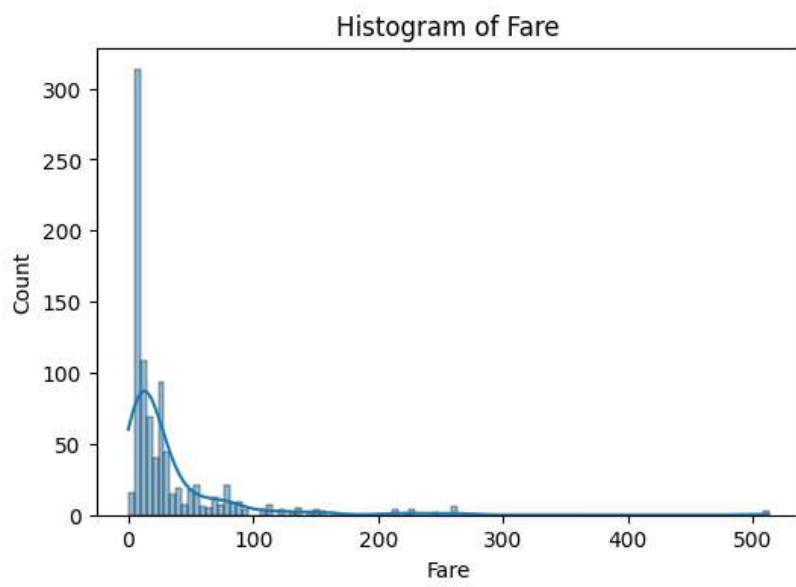
print("\nMissing Values After Cleaning")
print(df.isnull().sum())
```

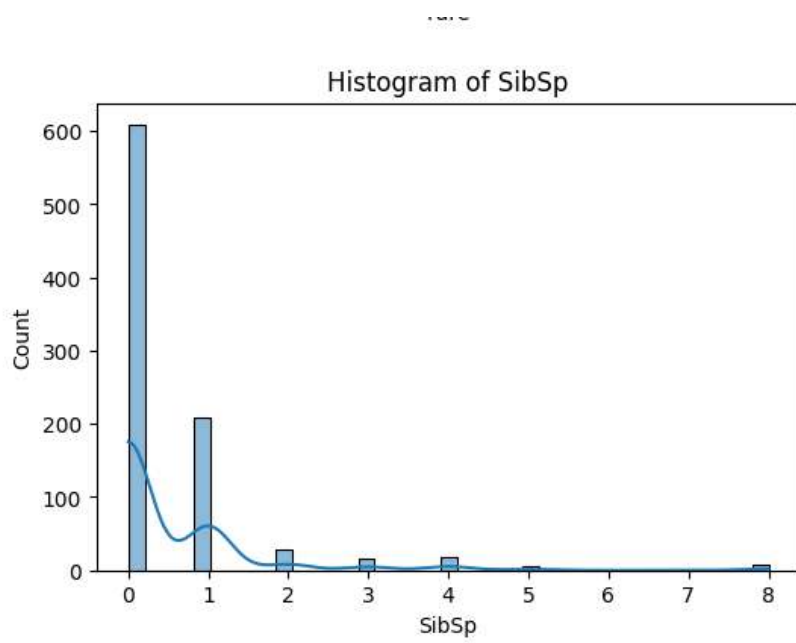
```
Missing Values After Cleaning
PassengerId    0
Survived        0
Pclass         0
Name           0
Sex            0
Age            0
SibSp          0
Parch          0
Ticket         0
Fare           0
Embarked       0
dtvpe: int64
```

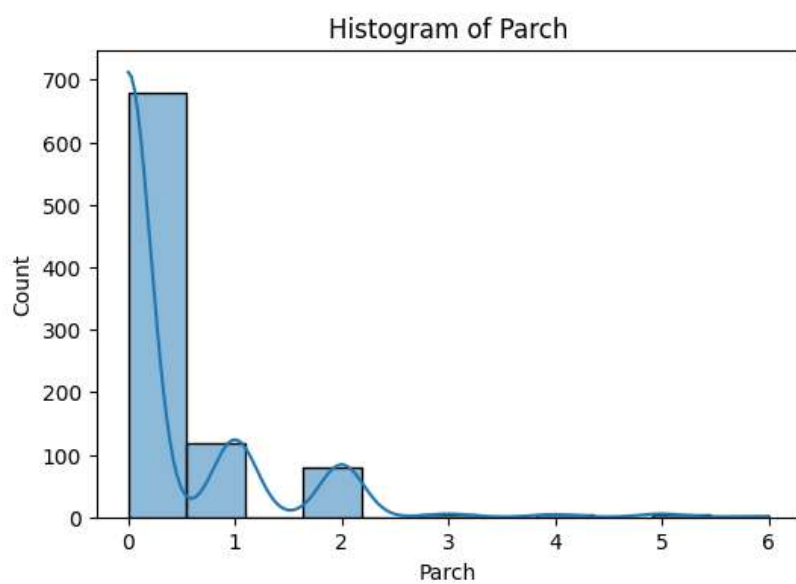
```
numeric_cols = ['Age', 'Fare', 'SibSp', 'Parch']
```

```
for col in numeric_cols:  
    plt.figure(figsize=(6,4))  
    sns.histplot(df[col], kde=True)  
    plt.title(f'Histogram of {col}')  
    plt.show()
```

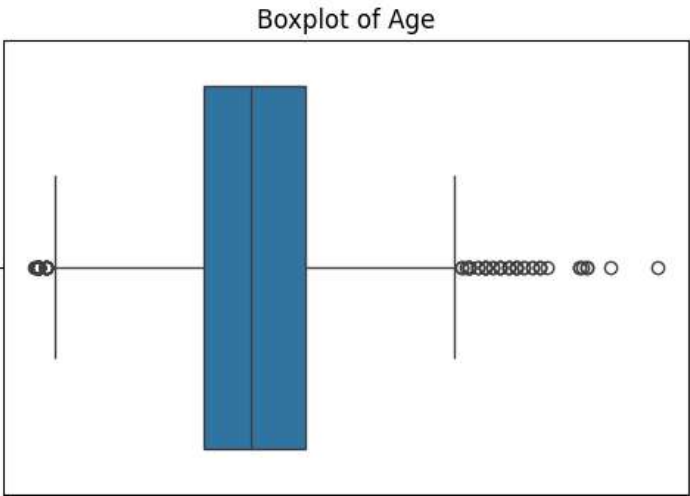






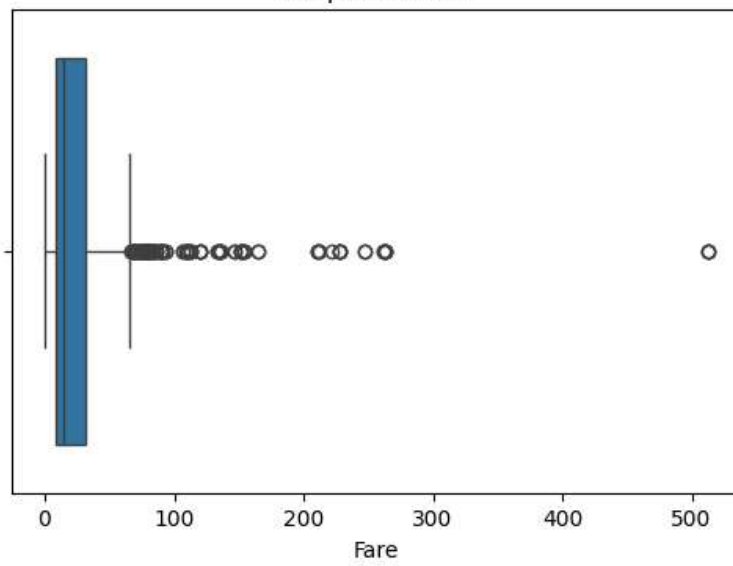


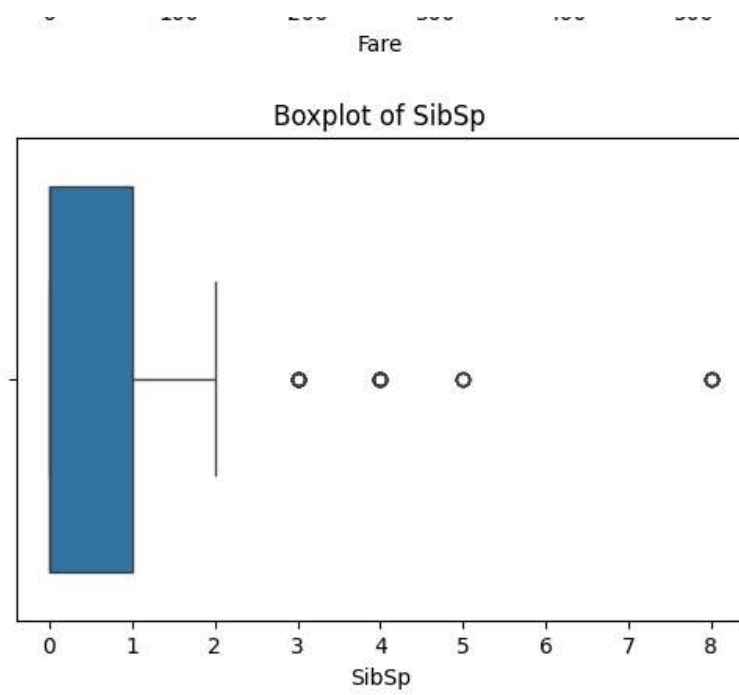
```
In [18]: # =====  
# BOXPLOTS  
# =====  
  
for col in numeric_cols:  
    plt.figure(figsize=(6,4))  
    sns.boxplot(x=df[col])  
    plt.title(f'Boxplot of {col}')  
    plt.show()
```



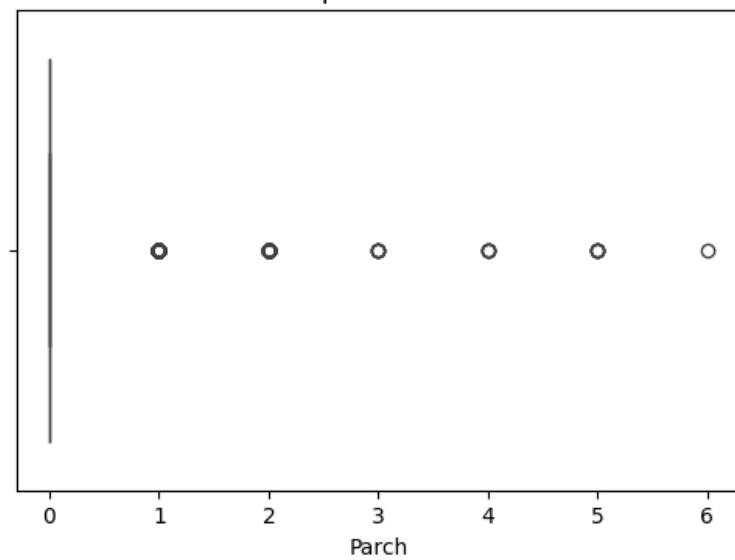
Age

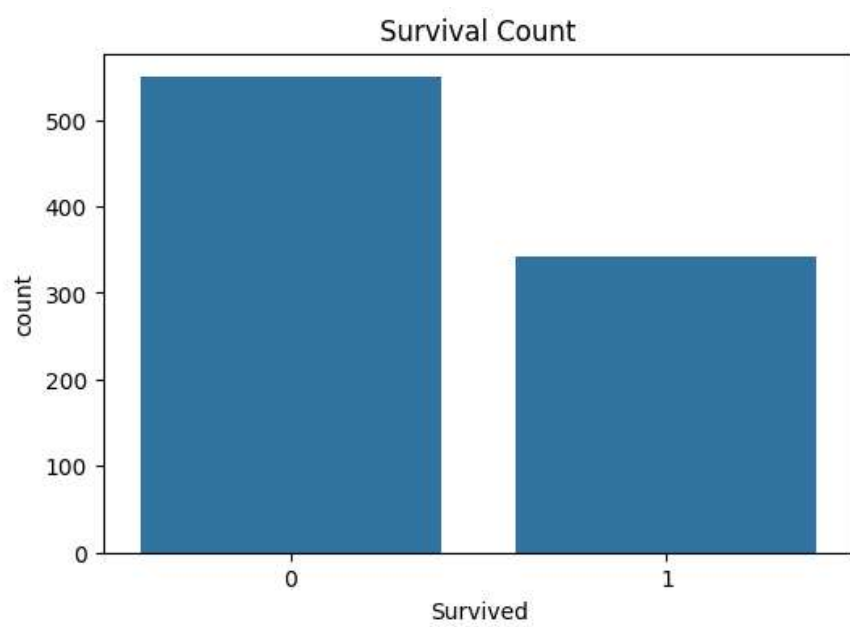
Boxplot of Fare

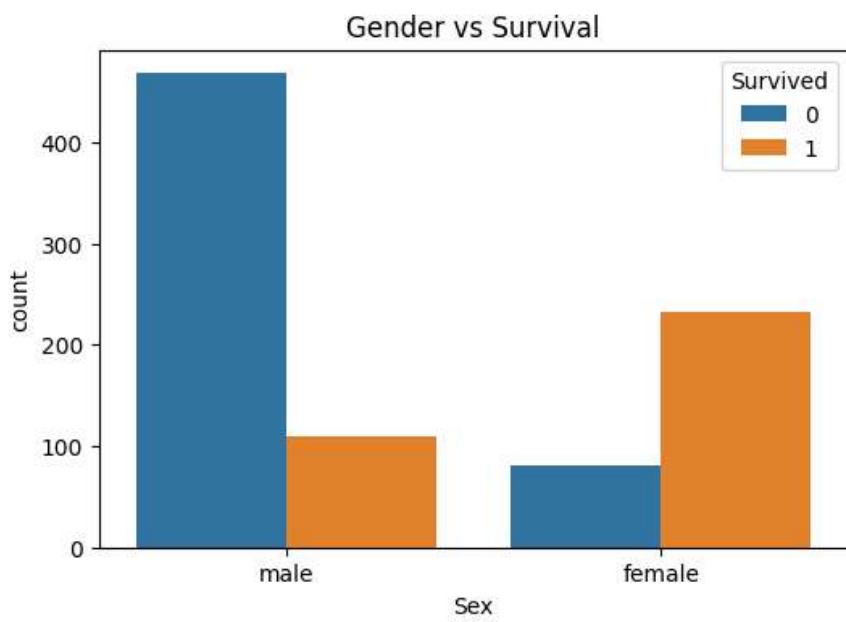


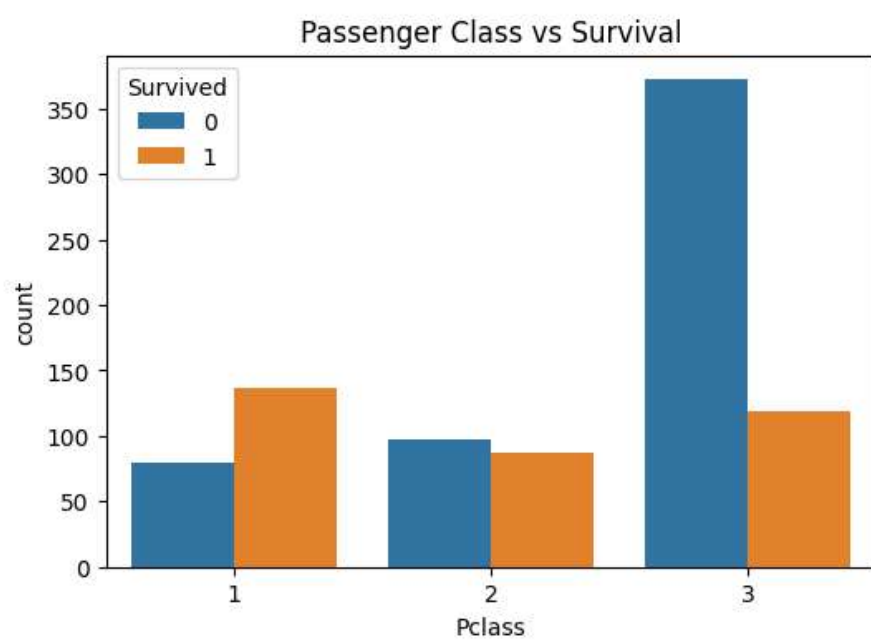


Boxplot of Parch

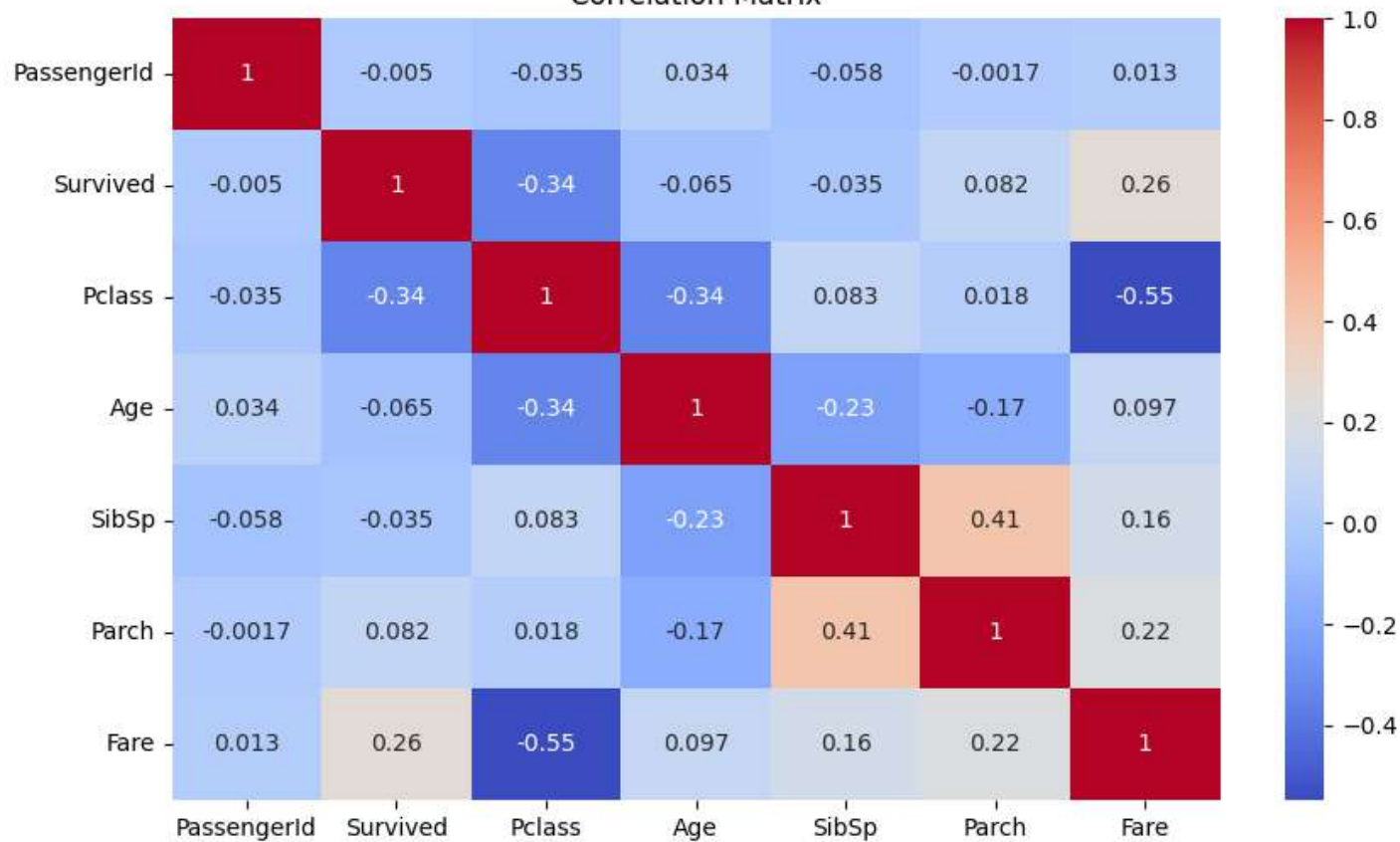


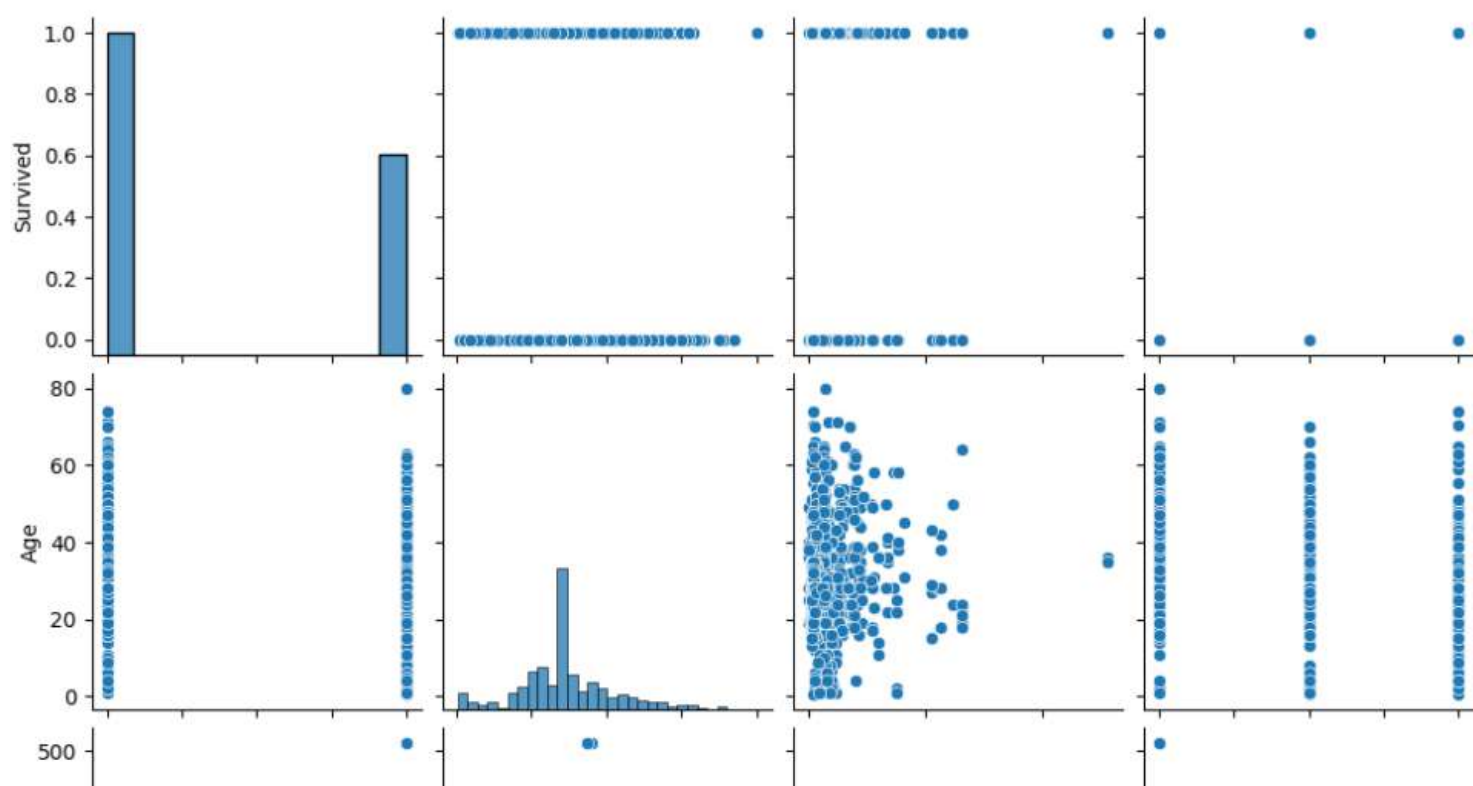






Correlation Matrix





```
Number of Fare Outliers:  
116
```

Key Insights

1. Females survived more than males.
2. Higher-class passengers had higher survival rates.
3. Fare contains significant outliers.
4. Age distribution is slightly right-skewed.
5. Fare and Pclass show correlation.

```
[ ]:
```